Sets — Chapter 1

A **set** is any well-defined collection of elements or members.

Describing a set

List the elements between braces (a finite number of elements), or specify a property that the elements of the set have in common:

$$R = \{1, 2, 3, 4\}$$
$$R = \{x : x \text{ is a real number}\}.$$

The second description is the **rule method**, used for infinite sets.

Order and repetition

The order of elements in a set is not important:

$$\{1, 2, 3\} = \{2, 1, 3\} = \{2, 3, 1\}.$$

Repeated elements are ignored:

$$\{1, 2, 3, 1\} = \{1, 2, 3\}.$$

Notation

Uppercase letters denote sets: $A, B, C, X, \ldots$. Lowercase letters denote elements: $a, b, c, x, \ldots$.

$x \in A$ $\quad$ x belongs to the set A,
$x \notin A$ $\quad$ x does not belong to the set A.

Fuzzy set

A set that is not well-defined (random elements put together), for example:

$$X = \left\{0, 4, -2, 7.5, -\frac{22}{7}\right\}.$$

Venn diagrams

Sets are represented as geometrical shapes for easier representation. Example:

$$A = \{1, 2, 3, 4\}, \qquad B = \{3, 4, 5, 6, 7\}.$$